

gdTAI V4.2 integration and pseudo-NK consensus QC

Execution date: 16 August 2026. This report audits the recovered development sidecar, A100 scVI fit, repeated RAPIDS clustering, and expression-independent pseudo-NK consensus. Locked validation cohorts were excluded throughout.

Scientific result: FAIL - no usable pseudo-NK cells

The computation completed correctly, but zero of 113,287 eligible cells satisfied the frozen consensus contract. Classifier fitting from this pseudo-label lane must not proceed.

4,023,462	21,054	3,978,014	0
development cells	primary NK anchors	boundary cells (98.9%)	selected pseudo-NK cells

Executive interpretation

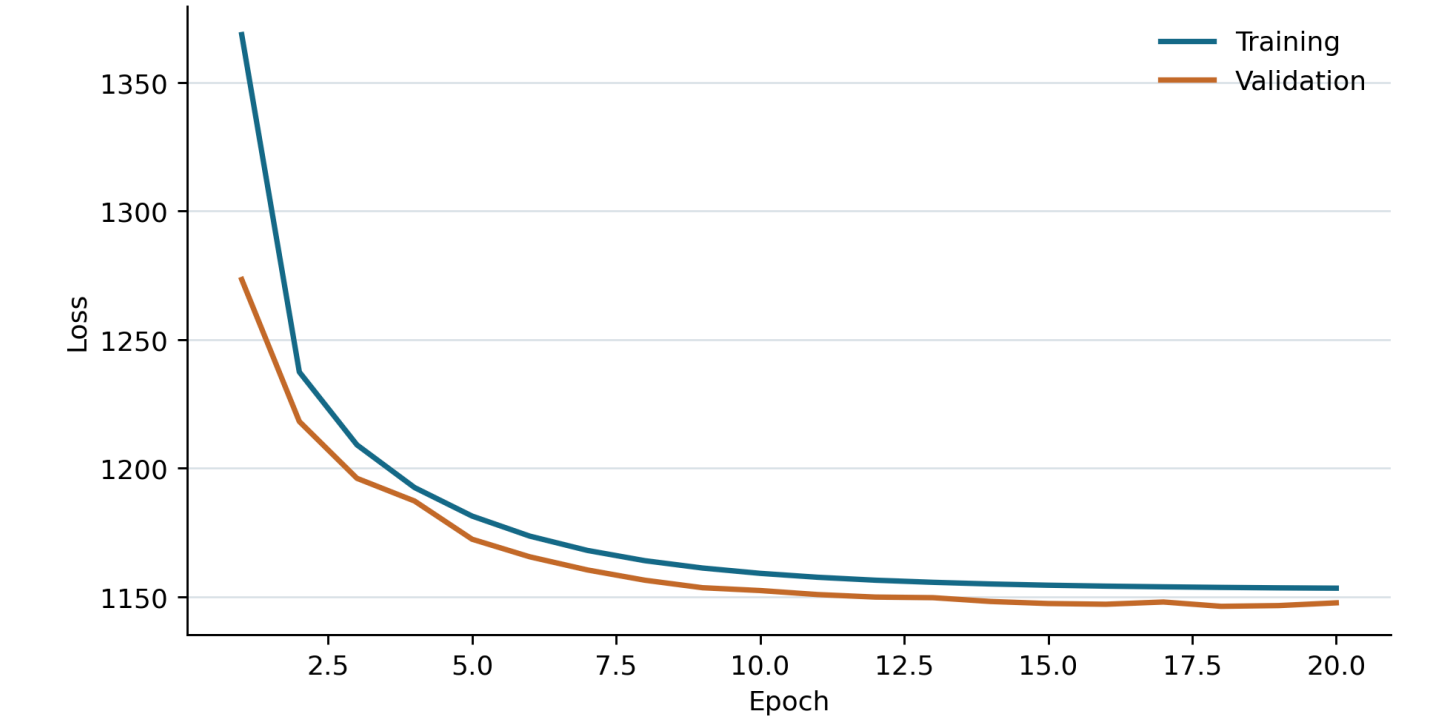
Staging, GPU integration, and clustering passed their technical contracts: the sidecar contains 4,023,462 cells by 4,000 common HVGs, scVI produced a finite 30-dimensional latent representation on the A100 without CPU fallback, and all 18 Leiden partitions were written with a verified SHA-256 checksum.

The pseudo-label contract failed scientifically. Six clusters reached at least 95% NK purity among independent anchors and no more than 2% productive-TCR contamination. Each also passed the anchor-count and multi-source-presence rules, but 87%-90% of its eligible candidate cells came from one dataset, exceeding the frozen 70% single-source cap. Therefore no cluster qualified in any run and no cell could receive the required 80% agreement.

Why this is not a classifier result. No gdTAI classifier, calibration model, decision threshold, release artifact, or whole-atlas inference was fitted in this execution. V2/V3 performance cannot be compared to an empty V4.2 training lane.

Integration QC

A100 scVI training history

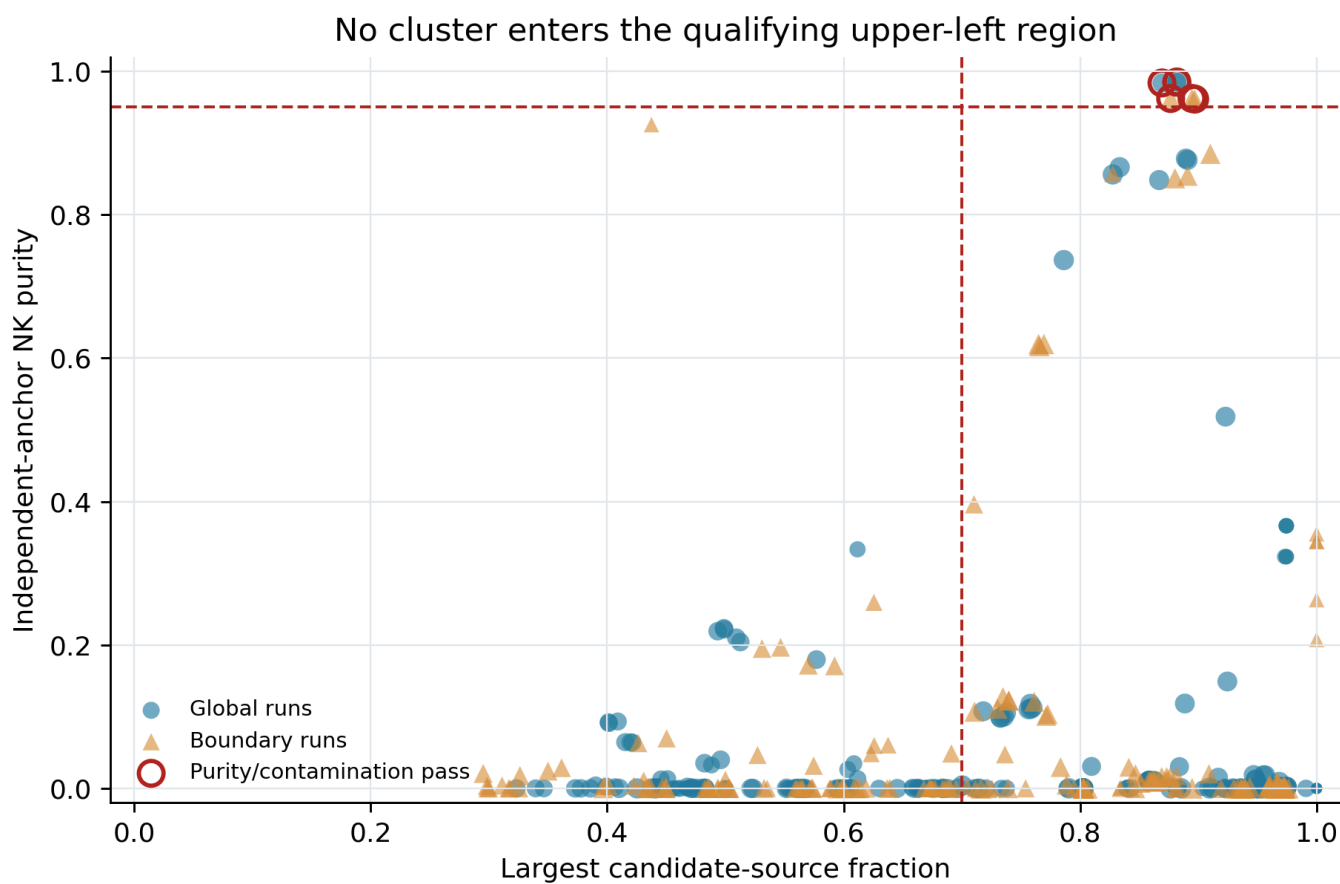


Training and validation loss across the frozen 20-epoch A100 scVI fit.

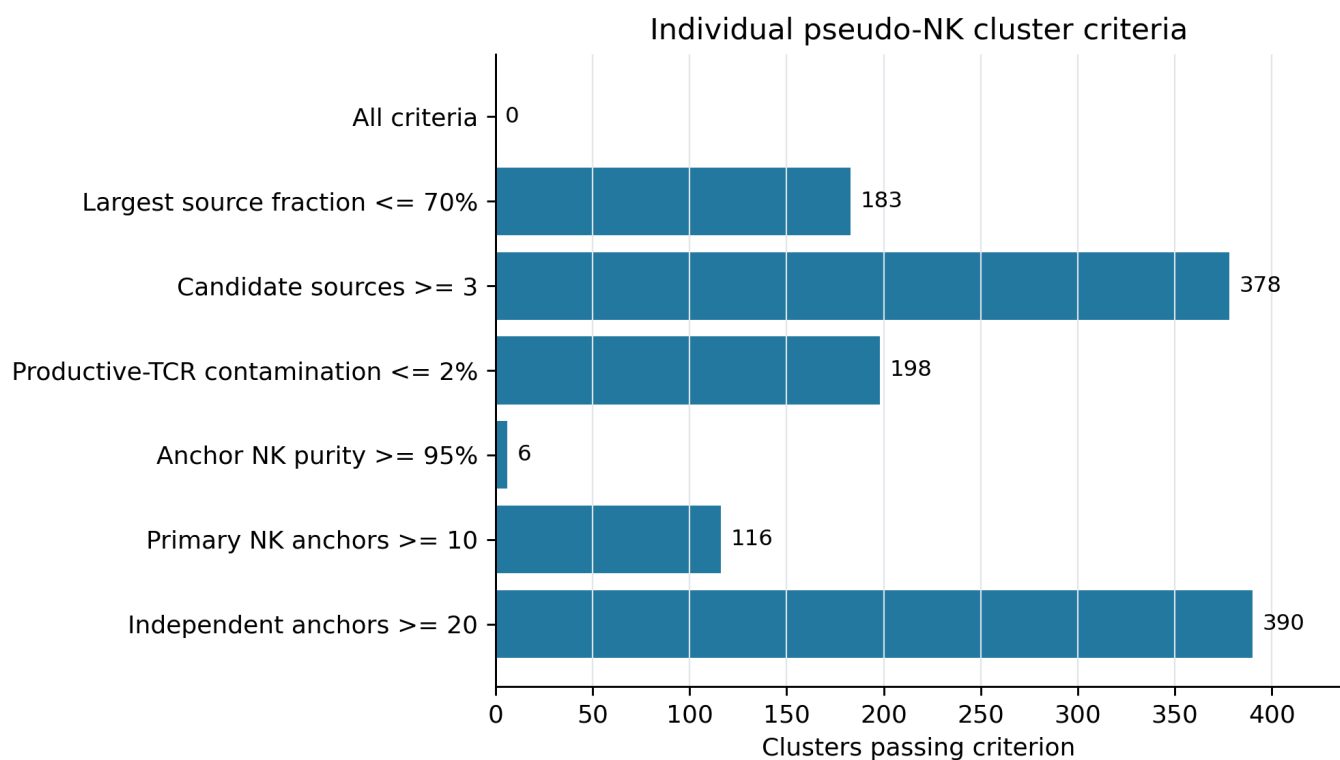
check	status	detail
Recovered sparse staging	PASS	4,023,462 cells x 4,000 HVGs
Locked cohorts excluded	PASS	Zero locked-cohort cells in staging and consensus
Source H5ADs unchanged	PASS	6/6 size/mtime pairs unchanged
A100 scVI fit	PASS	30 latent dimensions; 22.6 minutes
RAPIDS partitions	PASS	9 global + 9 boundary runs
Partition checksum	PASS	90e83ec83986358a015885e02e29d06eacf726d928e02b2e90428d5c09947a63
Boundary localization	WARN	3,978,014/4,023,462 cells (98.87%)
Qualifying clusters	FAIL	0/396 clusters
Selected pseudo-NK cells	FAIL	0/113,287 eligible cells
Classifier fitting	PASS	Not performed

Clustering and boundary QC

The expression-independent boundary was defined as the union of global clusters containing both a dual-annotation primary NK anchor and a productive-TCR anchor in at least one global run. It included 3,978,014 cells (98.87% of the sidecar), indicating that the global clusters were too broad for this union to localize the biological T/NK boundary.



Dashed lines mark the 95% anchor-purity and 70% maximum-source thresholds. The six high-purity clusters all fall to the right of the source-balance limit.



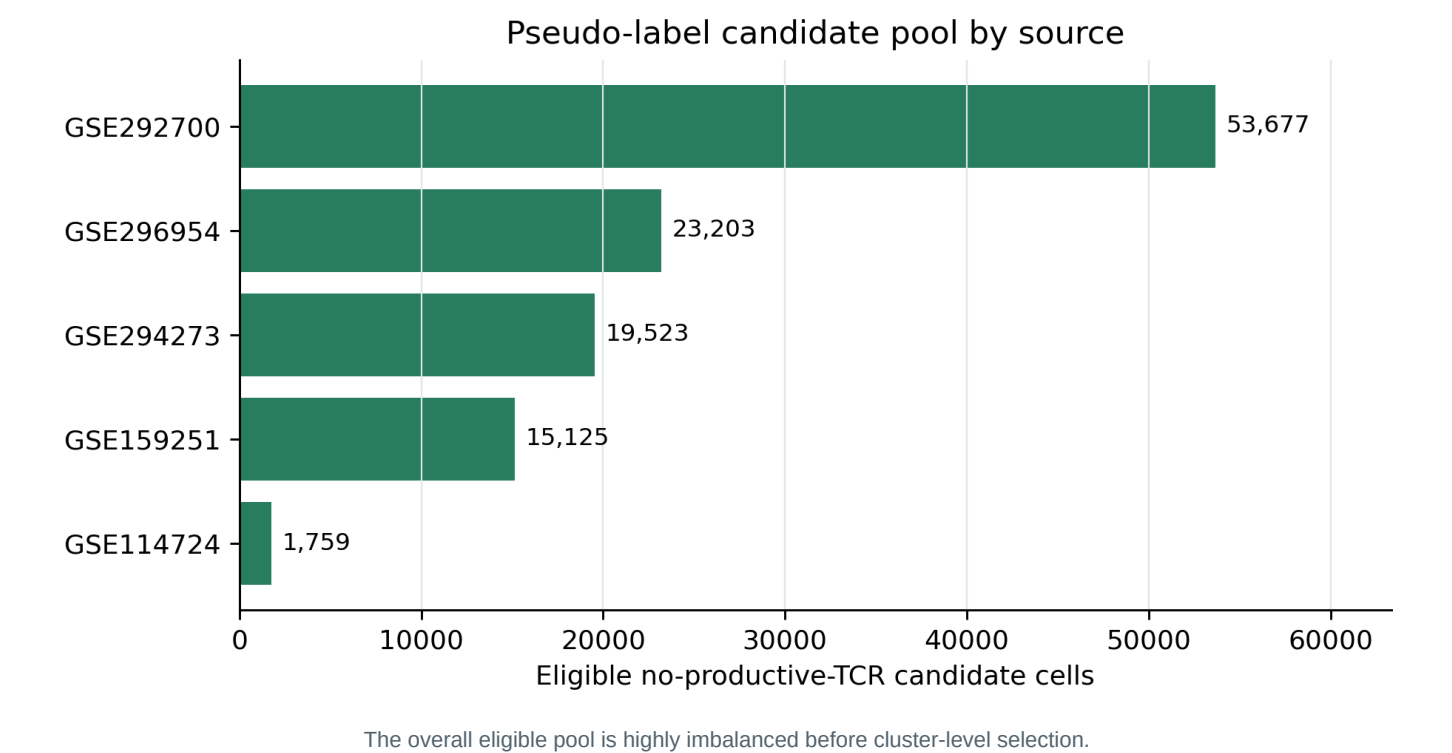
Individual criteria are feasible, but their frozen intersection contains no cluster.

Near-qualifying clusters

These are the only clusters that passed the frozen purity and productive-TCR contamination limits. Their sole failing field is `maximum_candidate_source_fraction`.

Run	Cluster	NK anchors	TCR anchors	NK purity	TCR/cell	Dominant source	Source share
global_r1.2_s11	12	19755	328	98.367%	0.102%	GSE292700	86.923%
global_r1.2_s29	12	19751	308	98.465%	0.098%	GSE292700	88.168%
global_r1.2_s47	10	17221	263	98.496%	0.118%	GSE292700	88.187%
boundary_r1_s11	18	20267	799	96.207%	0.202%	GSE292700	89.459%
boundary_r1_s29	16	20267	801	96.198%	0.199%	GSE292700	87.655%
boundary_r1_s47	19	20113	813	96.115%	0.211%	GSE292700	89.688%

Candidate-source composition



Dataset	Eligible candidates	Pool share	Selected pseudo-NK
GSE114724	1759	1.55%	0
GSE159251	15125	13.35%	0
GSE292700	53677	47.38%	0
GSE294273	19523	17.23%	0
GSE296954	23203	20.48%	0

Decision and next iteration

Do not fit the planned V4.2 classifier from these pseudo-labels. Relaxing the 70% cap after seeing this result would be outcome-driven tuning and would reintroduce the source-label shortcut the cap was designed to prevent.

The next defensible iteration should repair the reference design rather than lower the purity standard: source-balance the eligible candidate construction before graph inference; increase independent, confidently annotated NK anchors from multiple development datasets; use finer local neighborhoods or anchor-neighborhood propagation instead of a union of very broad Leiden clusters; and retain the locked cohorts unchanged for nested comparison against V2 and V3.

Reproducibility

- Execution config SHA-256: e70f1fc2f343011acb2f027a1a32fce2ec19467234e7236798bb42cdcaa16931
- Runner SHA-256: 65b319ec9bc7b73cd641b09e0f311ce7a57bc35ea62957524a0bd2de87c57659
- Core SHA-256: 191fe373beb827d7f8c63260ffbd066b1bbd08289318fbc23471eaddfc4b8a9
- Partition SHA-256: 90e83ec83986358a015885e02e29d06eacf726d928e02b2e90428d5c09947a63
- Report builder SHA-256: f5b9173afaa011b8534601928cebcf72813e81805dc4f110ea0a58a99e8c8b82
- Classifier fitting performed: **no**